

Total No. of Questions : 4]

SEAT No. :

P8578

[Total No. of Pages : 2

Oct-22 /TE/Insem-558
T.E. (Information Technology)
OPERATING SYSTEMS
(2019 Pattern) (Semester-I) (314442)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4.
- 2) Assume suitable data if necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right side indicate full marks.

- Q1) a)** Explain the role operating system as resource manager. [5]
- b) Give the significance of following shell commands with example: is, uniq, tail, touch grep. [5]
- c) Describe the differences between a monolithic kernel and a microkernel. [5]

OR

- Q2) a)** What is an operating system? List and explain services provided by the operating system. [5]
- b) Write a shell script to check if the given string is palindrome or not. [5]
- c) Explain about the concept of virtual machines and its advantages. [5]
- Q3) a)** Consider the Set of Processes with Arrival Time, Burst Time & Priority [8]

Process	Arrival Time	Burst Time
P1	7	5
P2	3	4
P3	10	3
P4	0	8
P5	12	6

P.T.O.

Find Average Turnaround Time & Average Waiting Time for SJF (Preemptive)& Round Robin (Time Quantum=2) scheduling algorithms with the help of Gantt chart

- b) With the help of neat, explain in detail process state transition diagram with two suspend states. [7]

OR

- Q4)** a) Discuss with the help of neat diagram different thread models. [5]
b) List and explain the CPU scheduling criteria. [5]
c) Explain with the help of neat diagram the process of context switching, also explain how program counter plays its role in context switching. [5]



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P5055

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**T.E. (Information Technology) (Insem)
OPERATING SYSTEMS
(2019 Pattern) (Semester - I) (314442) (Theory)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Assume suitable data if necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Describe the concepts of virtual machines with its implementation and benefits. Brief the example of virtual machine. **[7]**

b) Define operating system. Explain basic functions of operating system. **[8]**

OR

Q2) a) Explain the following operating systems **[5]**

i) Batch OS

ii) Real time OS

b) Illustrate the difference between a micro Kernel and a layered Kernel architectures of operating system with the help of neat diagrams. **[5]**

c) Write a shell script to find the factorial of a given number. **[5]**

Q3) a) Enlist different process states. Draw and explain process state transition diagram. **[7]**

P.T.O.

- b) Consider the following set of processes with the length of the CPU burst time given in milliseconds.

Process	Arrival Time	Burst Time	Priority
P ₁	0	3	2
P ₂	1	6	1
P ₃	4	4	3
P ₄	6	2	4

Draw the gantt charts illustrating the execution of these processes using priority (non preemptive) and SJF (preemptive) algorithms. Calculate average waiting time and average turnaround time. [8]

OR

- Q4)** a) Differentiate between user level threads and Kernel level threads. [5]
 b) Why PCB is required? Explain the elements of PCB. [5]
 c) Explain fork () and wait () system call in detail. [5]

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P5055

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[6187]-458

**T.E. (Information Technology) (Insem)
OPERATING SYSTEMS
(2019 Pattern) (Semester - I) (314442) (Theory)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Assume suitable data if necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Describe the concepts of virtual machines with its implementation and benefits. Brief the example of virtual machine. **[7]**

b) Define operating system. Explain basic functions of operating system. **[8]**

OR

Q2) a) Explain the following operating systems **[5]**

i) Batch OS

ii) Real time OS

b) Illustrate the difference between a micro Kernel and a layered Kernel architectures of operating system with the help of neat diagrams. **[5]**

c) Write a shell script to find the factorial of a given number. **[5]**

Q3) a) Enlist different process states. Draw and explain process state transition diagram. **[7]**

P.T.O.

- b) Consider the following set of processes with the length of the CPU burst time given in milliseconds.

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P ₃	4	4	3
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Draw the gantt charts illustrating the execution of these processes using priority (non preemptive) and SJF (preemptive) algorithms. Calculate average waiting time and average turnaround time. [8]

OR

- Q4)** a) Differentiate between user level threads and Kernel level threads. [5]
 b) Why PCB is required? Explain the elements of PCB. [5]
 c) Explain fork () and wait () system call in detail. [5]

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SEAT No. :

PC55

[Total No. of Pages : 2

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T.E. (Information Technology) (Insem)

OPERATING SYSTEMS

(2019 Pattern) (Semester - I) (314442)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) What is multiprogramming, multitasking and multiprocessing? Discuss with an example. **[5]**

b) Define operating system. Discuss objectives and functions of operating system. **[5]**

c) Discuss following command with an example - pwd, ps, touch, fork, uniq. **[5]**

OR

Q2) a) Explain serial processing OS and simple batch OS. **[5]**

b) Write a shell script to identify whether the given string is palindrome or not. **[5]**

c) Describe the read and echo commands with various options in BASH scripting. Provide an example of how you can use it. **[5]**

P.T.O.

Q3) a) How does a process move between different states in an operating system. Explain with the help of state transition diagram. [7]

b) Consider a multitasking OS, with four tasks, with the length of the CPU burst time given in milliseconds.

Process	Arrival Time	Burst Time	Priority
1	0	3	4
2	2	6	2
3	4	4	3
4	5	2	6

Draw the Gantt chart. Using SJF (non preemptive) and priority (preemptive). Scheduling algorithms. Find average WT and average TAT also. [8]

OR

Q4) a) Illustrate the difference between process and thread. Explain the typical entries in PCB and thread control block. [7]

b) Explain the various types of processor schedules with the help of process state transitions. [8]

